



DebugThugs



Problem Statement: Smart Campus Printing & Document Service

Team Name: Debug Thugs

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Description: DocPrint is a smart digital printing platform that connects students with nearby printing shops for seamless document upload, customization, ordering, and delivery. It reduces queues and delays through real-time tracking, QR pickup, printer monitoring, and intelligent order rerouting.

1. Printing Shop Registration & Service Marketplace

Shop Owner Registration

Printing shops and document service providers can **create an account** and join the **DocPrint platform**.

They can list their shop, making it easier for students to **discover, compare**, and choose the best printing center based on **location, price, speed, and service quality**.

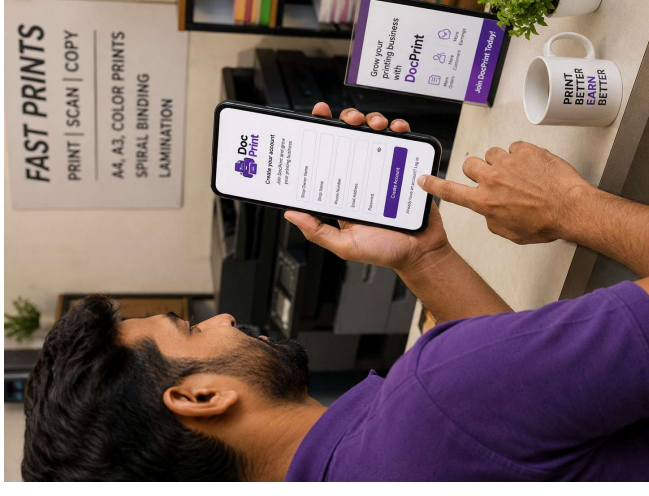
Service Listing

Each printing center can clearly specify the services they provide, such as:

Black & White • Color Printing • Lamination • Photo Printing • Large-Format Printing • 3D Printing

Key Benefits

-  Creates a centralized printing ecosystem
-  Shows users only shops capable of fulfilling their requirements
-   Helps local printing businesses reach more customers
-  Encourages competition through better pricing, speed, and service quality



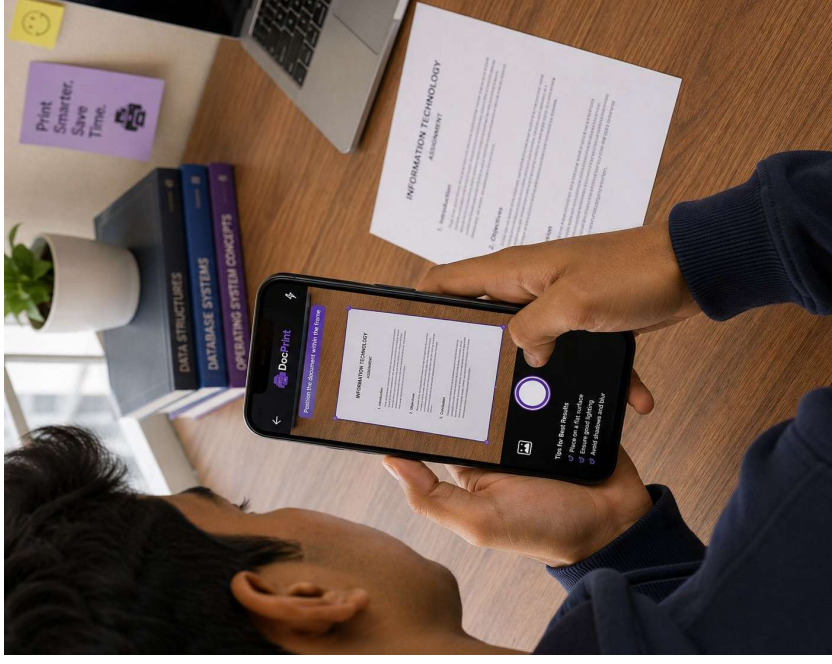
2. Users upload document in App

Users can configure:

- Number of copies
- Page range
- Desired delivery time
- Color or black & white
- Single or double-sided printing
- Paper size
- Binding options

Benefit

Students no longer need to stand in a queue simply to submit a document.



3. Automatic Document Generation

Generate Documents Directly by discussing with AI.

or

The platform can provide document templates for common student requirements.

Examples:

- Leave application
- Assignment cover page
- Project report cover page
- Bonafide request
- Internship letters
- Certificates
- Formal letters

Users just need to fill in required information, and the system automatically generates the document.

4. Document Editing Before Printing

Edit Documents Without Leaving the Platform

Users can make basic modifications before printing.

Possible features:

- Edit text
- Add images
- Change formatting
- Add signatures
- Merge documents
- Split documents
- Reorder pages

This reduces the need to use multiple applications before printing.

5. Smart Document Upload & Automatic Generation

Upload & Customize Documents

Users can upload documents directly through the DocPrint app and configure:

Copies • Page Range • Delivery Time • Color/B&W • Single/Double-Sided • Paper Size • Binding Options

Benefit: Students can submit printing requests digitally without standing in a queue.

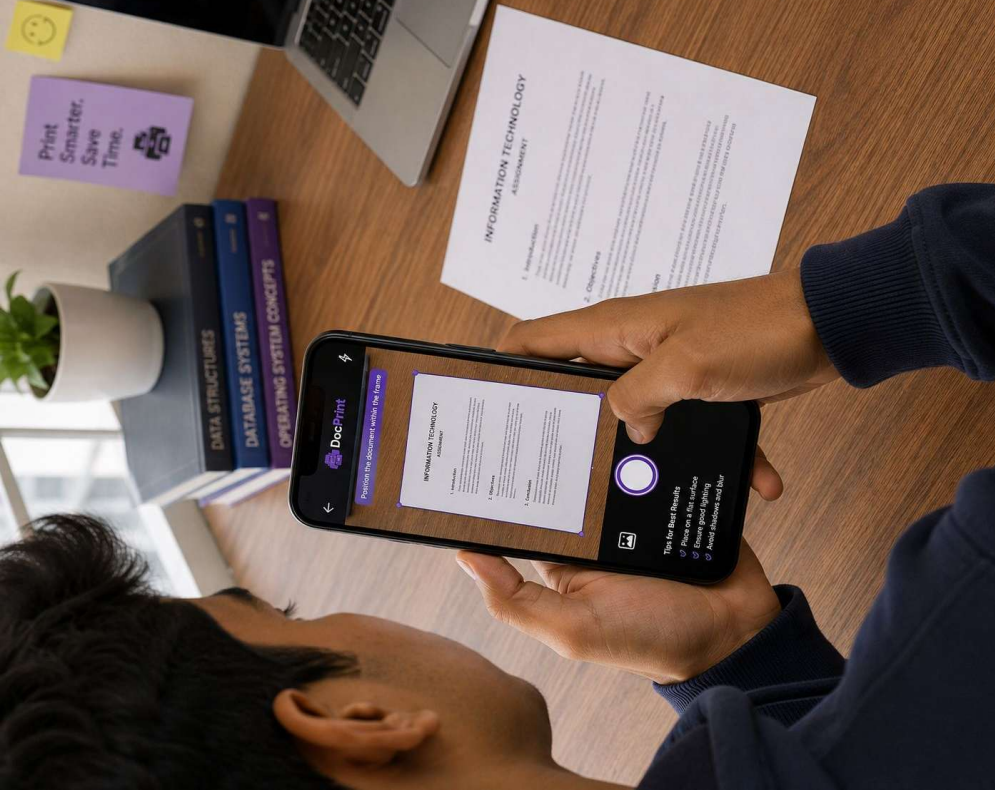
Automatic Document Generation

DocPrint also provides templates for commonly required student documents, such as:

Leave Applications • Assignment Cover Pages • Project Report Covers • Bonafide Requests • Internship Letters • Certificates • Formal Letters

Users simply **fill in the required details**, and the platform automatically generates a ready-to-use document.

Result: Create + Customize + Print — All in One App



6. Reusable Student Document Library

Frequently Used Campus Documents

The platform provides a library of reusable templates and commonly required documents.

Examples include:

- Resume templates
- Assignment formats
- Lab record formats
- Project report templates
- College application forms
- Cover pages

Students can select a template, customize it, and directly send it for printing.

Benefits

- Saves time
- Maintains standardized formats
- Reduces repetitive work

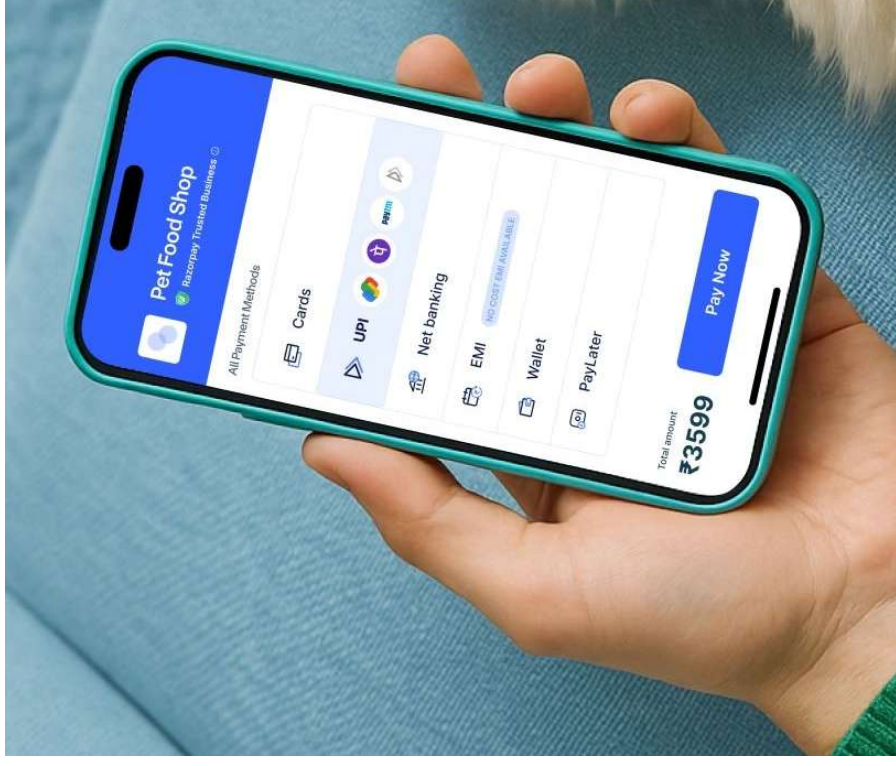
7. Secure Digital Payments

Razorpay & UPI Integration

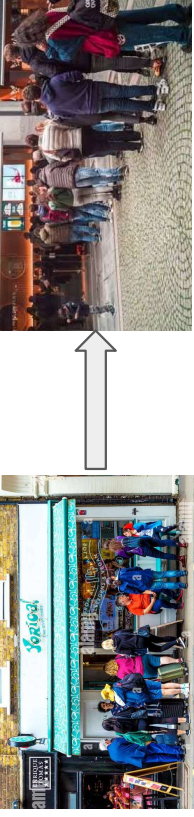
The platform integrates digital payment systems such as **Razorpay and UPI**.

Users can:

- Pay before printing
- Receive payment confirmation
- View transaction status
- Access order and payment history



8. Smart Queue Management



“If there two orders came. First is of 50 pages and second is 3 pages. Then 3 pages print should be processed first , to save time”

Instead of processing documents simply in the order they arrive, the system intelligently manages the printing queue.

The platform estimates factors such as:

- Number of pages
- Number of copies
- Current queue length
- Printer availability
- Estimated printing duration

Based on these factors, the system calculates the most efficient order in which jobs should be processed.

Benefits

- Reduces waiting time
- Improves printer utilization
- Prevents printer overload
- Provides more accurate completion estimates
- Reduces unnecessary crowd in front of shops

9. Estimated Printing Time



- Users can see the estimated time in their app
- The estimation considers:
 - Current printing queue
 - Number of pages
- The estimate can dynamically update if the queue changes or printer availability changes.



10. Queue Prediction

Predicting Future Waiting Time

The system analyzes current printing activity to predict:

- Expected waiting time
- Busy periods
- Printer workload
- Potential delays

This helps users decide:

- Which printing center to select
- Whether to wait or choose another center
- When their document is likely to be ready

The platform can eventually use historical data and machine learning to improve prediction accuracy.

Incoming Orders

8

+3 new from yesterday

Active Printing Queue

12

Orders in progress

Total Revenue

₹18,750

+22% from yesterday

Completed Orders

36

Today

New Customers

14

+5 from yesterday

Active Printing Queue

Order ID	Customer	Service	Pages	Status	Est. Time
#DP3256	Karan Patel	B&W Print	12	Printing	10 min
#DP3257	Riya Sharma	Color Print	8	Printing	15 min
#DP3258	Ankit Verma	Project Report	25	In Progress	20 min
#DP3259	Neha Singh	Lamination	2	Pending	25 min
#DP3260	Aditya Jain	B&W Print	30	In Progress	18 min

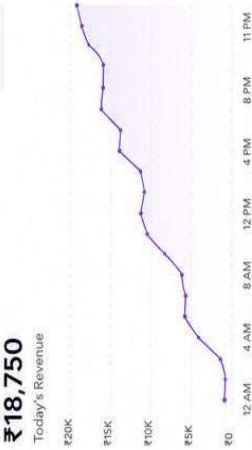
View Full Queue

Printer Status

Printer 1 (B&W)	Ready	72%
Printer 2 (Color)	Printing	45%
Printer 3 (Large Format)	Idle	20%
Printer 4 (3D Printer)	Offline	0%

Manage Printers

Revenue Tracking



View Detailed Report

Customer Analytics

New vs Returning Customers



View All Customers

Popular Services

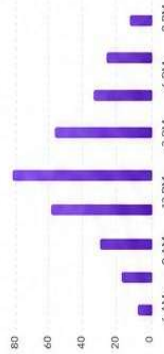
Most Ordered Services

B&W Printing	126 orders
Color Printing	78 orders
Lamination	45 orders
Spiral Binding	32 orders
Project Report	30 orders

View Services

Peak Hours

Busiest Time Slots



Detailed Analytics

Order History

Recent Completed Orders

#DP3250	Rohan Mehta	Completed
#DP3249	Priya Kumari	Completed
#DP3248	Siddharth Rai	Completed
#DP3247	Megha Gupta	Completed
#DP3246	Arijun Nair	Completed

View All Orders

Printer Utilization

Overall Printer Usage Today

59%

Average Utilization

12. Intelligent Printing Price Estimation

Instead of using only a fixed price per page, the system can calculate an estimated printing cost based on document characteristics.

Possible factors include:

- Number of pages
- Color coverage
- Ink usage estimation
- Paper size
- Paper type
- Number of copies
- Printing method

For image-heavy documents, the system can estimate that higher ink consumption may increase the printing cost.

13. Automatic Printer Failure Detection & Smart Order Rerouting

Real-time monitoring • Instant alerts • Automatic rerouting or refund

1. Printer Health Monitoring

The system continuously monitors the health of all connected printers.

Printer Online
Shop A – Printer 1

Paper Shortage
Shop B – Printer 2

Ink Shortage
Shop C – Printer 1

Printer Error
Shop D – Printer 3

Connection Failure
Shop E – Printer 2

Hardware Malfunction
Shop F – Printer 1

Printer Status Overview

14 Online

4 Warning

3 Error

3 Offline

Recent Alerts

Shop D – Printer 3
Printer Error
2 min ago

Shop B – Printer 2
Paper Shortage
5 min ago

Shop C – Printer 1
Ink Shortage
10 min ago

Shop Owner Dashboard

Overview

Orders

Printers

Alerts

Reports

Settings

Alerts

Mark all as read

Printer Error
Shop D – Printer 3 is not working properly.
2 min ago

Paper Shortage
Shop B – Printer 2 is running low on paper.
5 min ago

Ink Shortage
Shop C – Printer 1 is running low on ink.
10 min ago

Connection Failure
Shop E – Printer 2 is offline.
12 min ago



2. Instant Alerts

Shop owners and administrators receive real-time alerts.



3. Automatic Order Rerouting or Refund

If a printer becomes unavailable, the system automatically finds the best alternative or provides a refund based on user preference.

Original Order

Order ID: #DP3258

Color Printing – 50 Pages

Shop D – Printer 3

Failed

Printer Failed

Reason: Printer Error

Automatic Rerouting

The system finds the best available printer and reroutes the order.

Refund Option

If no suitable printer is available or user chooses refund.

Finding Best Alternative...

Distance

Printer Availability

Queue Length

Printing Capability (Color)

Estimated Completion Time

Best Match Found

Shop B – Printer 1

Available

Distance: 1.2 km

Est. Completion: 25 min

Queue Length: 2 orders

No Suitable Printer Found

No available color printer nearby can complete the order on time.

Refund Processed

Amount: ₹210

Refund Method: Original Payment

Status: Success

Order Rerouted

Your order has been rerouted successfully!

New Estimated Completion

25 min

Benefits

Minimize Delays

Smart Automation

Better Reliability

Happy Customers

How It Works



14. Smart Pickup & Delivery Experience

QR-Based Pickup

Choose what works best for you. Fast, secure and transparent.



Self Pickup

Get notified when the document is ready and collect it from the shop.



Order ID: #DP3258
Shop: Perfect Prints

1. Get Notified



2. Show QR Code



3. Verify Order



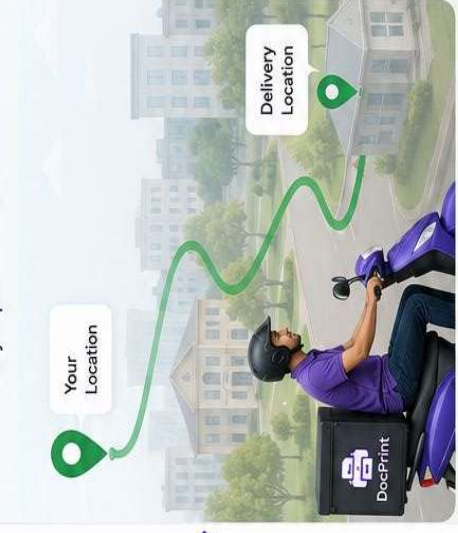
4. Collect Document

Faster Pickup



Delivery

Get documents delivered directly to hostels, academic buildings, libraries, or any requested location.



DocPrint

Your Location

Delivery Location

Hostels

Academic Buildings

Libraries

Any Location

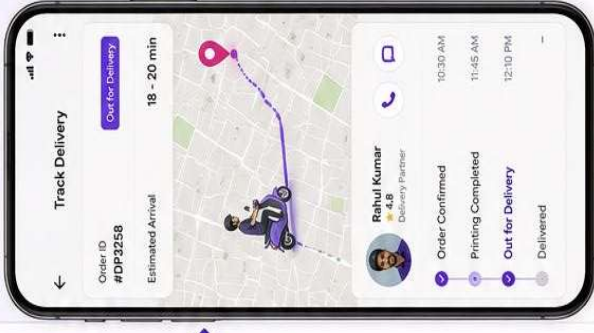
Secure & Contactless

Transparent Tracking



Real-Time Delivery Tracking

Track your delivery in real-time for a transparent and reliable experience.



Order ID: #DP3258

Estimated Arrival: 18 - 20 min

Live Delivery Status: Know exactly where your order is.

Delivery Partner Location: Track your delivery partner in real-time on the map.

Estimated Arrival Time: Get accurate ETA so you can plan your time better.

Better Experience

15. Privacy and Document Security

Your documents are safe with DocPrint

1. Upload Document

Student uploads document securely



2. Secure Transmission

Documents are encrypted during transmission



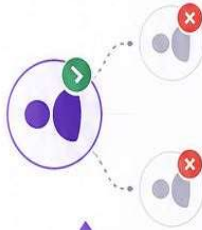
3. Secure Storage

Files are stored securely on encrypted servers



4. Authorized Access

Only authorized personnel can access documents



5. Print & Complete

Document is printed and order is completed



6. Automatic Deletion

Documents are automatically deleted after completion



End-to-End Security

From upload to deletion, your privacy is our priority.



Trusted & Secure

We follow industry best practices to keep your data safe.



Secure File Storage

Documents are stored in encrypted servers with multiple security layers.



Encryption in Transit

All documents are encrypted during transmission using industry-standard protocols.



Access Control

Role-based access ensures that only authorized staff can view documents.



Temporary Access

Documents are accessible only for the required time to complete the order.



Automatic Deletion

Documents are automatically deleted after completion to ensure maximum privacy.



Audit Logs

All activities are logged to monitor access and ensure full transparency.



Your Privacy. Our Priority.

We protect your documents like they're our own.

Tech Stack

- Frontend
 -  NEXT.js + Shadcn 
- Backend
 -  node + Express  js
- DB
 -  PostgreSQL + Drizzle ORM 
- Agentic
 -  LangChain +  Langgraph